

A Proximity to modeled SOL, T_SOL / T_k (GM over cases)

B Change in SOL proximity

Point-wise

Reduction/Norm.

MatMul/Attention

Stencil/Conv.

Data Layout

	B200			GH200			MI300X	Triton GH200 vs. B200	Triton MI300X vs. B200	cuTile GH200 vs. B200	TileLang GH200 vs. B200
	Triton	cuTile	TileLang	Triton	cuTile	TileLang	Triton				
dequantize_rowwise	0.34	0.29	0.33	0.46	0.41	0.45	0.39	1.39	1.17	1.42	1.37
dropout	0.73	0.73	0.74	0.81	0.79	0.81	0.59	1.10	0.80	1.08	1.10
fused_activation	0.83	0.82	0.82	0.88	0.87	0.87	0.60	1.07	0.73	1.07	1.07
leaky_relu	0.81	0.82	0.80	0.89	0.87	0.89	0.67±	1.09	0.82	1.07	1.10
mul2	0.64	0.61	0.66	0.75	0.72	0.75	0.59±	1.17	0.92	1.17	1.13
quantize_global	0.73	0.73	0.71	0.80	0.79	0.79	0.58	1.09	0.79	1.08	1.12
relu	0.63	0.59	0.65	0.74	0.72	0.74	0.59	1.17	0.94	1.22	1.15
rope	1.05	0.92	1.05	0.99	0.95	0.99	0.68	0.94	0.65	1.04	0.94
sigmoid	0.81	0.62	0.79	0.88	0.81	0.87	0.63	1.09	0.78	1.31	1.10
swiglu	0.90	0.73	0.84	0.94	0.84	0.93	0.69	1.05	0.77	1.15	1.10
vector_add	0.69	0.69	0.71	0.77	0.76	0.77	0.58±	1.12	0.84	1.10	1.09
weight_dequant	0.61	0.35	0.67	0.76	0.63	0.77	0.62±	1.24	1.01	1.78	1.15
argmax	0.43	0.45	0.52	0.60	0.57	0.63	0.43	1.38	1.00	1.28	1.21
batch_normalization	0.43	0.39	0.49	0.62	0.54	0.61	0.33	1.45	0.76	1.41	1.23
cross_entropy	0.18	0.13	0.17	0.29	0.22	0.30	0.17	1.57	0.91	1.73	1.76
histogramming	0.031	0.019	0.12	0.055	0.034	0.24	0.020	1.81	0.64	1.78	1.95
kl_divergence	0.46	0.48	0.51	0.61	0.60	0.60	0.44	1.32	0.96	1.24	1.18
l2_norm	0.67	0.59	0.71	0.73	0.74	0.79	0.58	1.08	0.86	1.26	1.11
layernorm	0.76	0.63	0.88	0.99	0.94	1.07	0.74	1.30	0.98	1.50	1.23
mean_reduction	0.60	0.53	0.61	0.64	0.63	0.64	0.47	1.05	0.78	1.19	1.05
moe_topk_gating	0.061	0.031	0.059	0.10	0.052	0.10	0.083	1.67	1.36	1.68	1.79
rmsnorm	0.87	0.82	0.97	1.03	1.03	1.14	0.79	1.18	0.90	1.27	1.17
softmax	0.52	0.34	0.46	0.65	0.52	0.63	0.43	1.27	0.83	1.54	1.37
batched_matmul	0.24	0.21	0.18	0.35	0.31	0.30	0.066	1.49	0.28	1.46	1.67
block_sparse_attention	0.18	0.067	0.067	0.30	0.14	0.18	0.042	1.67	0.23	2.07	2.65
flash_attention	0.49	0.53	0.29	0.61	0.70	0.45	0.14	1.22	0.27	1.34	1.52
flash_decode	0.007	0.002	0.008	0.016	0.005	0.016	0.005	2.17	0.72	2.37	2.18
linear_self_attention	0.009	0.005	0.003	0.017	0.014	0.006	0.005	1.87	0.58	3.07	2.14
matmul_fp32_fp16_fp8	0.90	1.14	0.54	0.99	0.62	0.41	0.078	1.10	0.09	0.54	0.77
matmul_int8	0.18	0.18	0.19	0.56	0.28	0.29	0.074	3.08	0.40	1.58	1.49
streamk_matmul	0.53	0.26	0.34	0.52	0.36	0.45	0.070	0.97	0.13	1.39	1.32
1d_conv	0.10	0.057	0.018	0.14	0.12	0.025	0.31	1.43	3.05	2.03	1.40
2d_conv	0.039	0.025	0.010	0.054	0.037	0.011	0.11	1.39	2.83	1.47	1.14
2d_max_pooling	0.34	0.26	0.43	0.51	0.49	0.45	0.36	1.51	1.08	1.88	1.05
3d_conv	0.047	0.022	0.012	0.068	0.033	0.015	0.20	1.44	4.16	1.51	1.34
gaussian_blur	0.12	0.10	0.089	0.15	0.12	0.16	0.14	1.20	1.13	1.17	1.77
jacobi_stencil_2d	0.36	0.41	0.42	0.55	0.60	0.62	0.39±	1.55	1.10	1.47	1.45
bitonic_sort	0.001	0.001	0.001	0.002	0.002	0.002	0.001	1.61	0.87	1.49	1.63
destindex	0.65	0.31	0.61	0.80	0.49	0.75	0.56	1.23	0.86	1.58	1.23
interleave	0.76	0.75	0.73	0.85	0.83	0.81	0.63	1.11	0.83	1.11	1.10
matrix_copy	0.45	0.46	0.45	0.54	0.53	0.54	0.44	1.20	0.97	1.15	1.20
matrix_transpose	0.78	0.78	0.63	0.88	0.88	0.78	0.63	1.13	0.80	1.12	1.23
radix_sort	0.017	0.009	0.010	0.027	0.015	0.015	0.022	1.56	1.25	1.58	1.58
reverse_array	0.55	0.54	0.47	0.71	0.67	0.62	0.55	1.29	1.01	1.24	1.34
top_k_selection	0.002	0.002	0.002	0.003	0.003	0.003	0.003	1.36	1.59	2.01	1.50



± MI300X bf16 cases use the conditional memory-dominance target (no calibrated BF16 vector peak)